

Cognitive Vulnerabilities in Heatwaves: Addressing Health Risks for At-Risk Populations

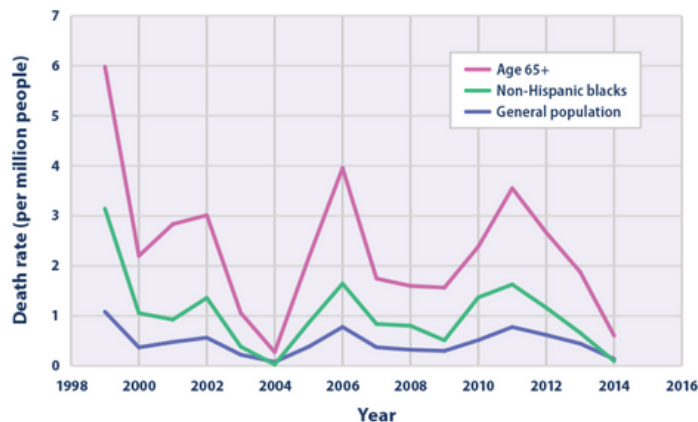
NOVEMBER 2024

This brief analyzes the inequities and barriers cognitively impaired individuals—such as dementia patients, medication users, and illicit drug users—face when receiving care for heat related illness. Based on the evidence from peer reviewed journals and government health data, this brief provides actionable strategies to prevent heat illness and provide adequate care for cognitively impaired individuals.

HEATWAVE HEALTH IMPACTS

From 2018 to 2022, heat related deaths surged by nearly 700, climbing from 1008 to 1714 fatalities.¹ The number of heat-related hospitalizations was averaging 9,235, with over 67,000 ED visits.² Heat illness complications including kidney failure or heart problems.³ Despite this increase, these numbers are likely “vastly” underreported since many heat related deaths are recorded as Cardiovascular disease—obscuring the true impact.⁴ Among those at risk are dementia patients— including the 6.5 million Alzheimer’s patients—who are often above 65— are 6% more likely to die on a hot day.⁵ Older adults are significantly more likely to die of heat related disease. See figure 1. With climate change increasing the frequency and intensity of future heat waves, these vulnerable populations face escalating risks for heat related illnesses. Without systemic intervention, heat-related deaths are projected to double by 2050.⁶

FIGURE 1: SUMMER DEATHS DUE TO HEAT IN THE UNITED STATES⁷



POPULATIONS AT RISK AND EMERGENCY ROOM CHALLENGES

Individuals with cognitive impairments or those who struggle to recognize or articulate signs of heat-related illnesses are more susceptible during extreme heat events. Cognitive impairments often

overlap with heat illness symptoms, which can lead to misdiagnosis and delays of treatment.⁸

Populations at risk:

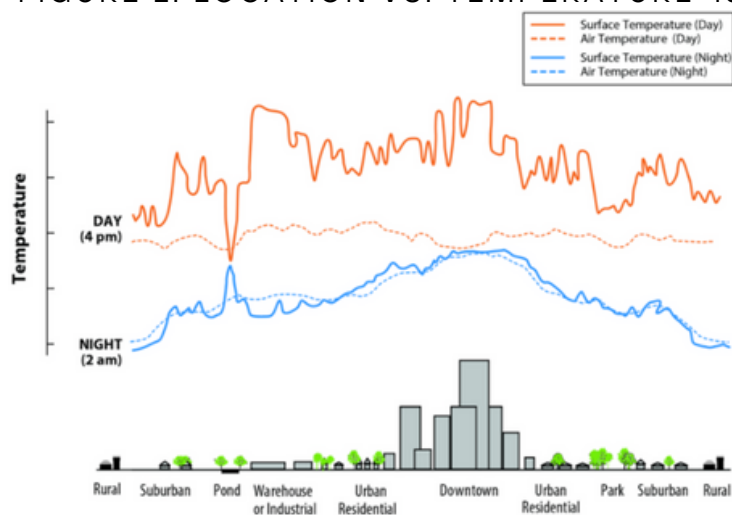
Individuals with dementia: Over 6.5 million Americans with Alzheimer’s depend on caregivers to recognize and articulate symptoms of heat illness.⁵ However, the shortage of caregivers and the lack of standardized training about early heat illness symptoms can result in delayed care and heightened chances of serious health complications.⁹

Additional characteristics that may place an Alzheimer’s patient at further risk includes:

1. Taking medications such as SSRIs, antipsychotics, or benzodiazepines— impairing cognitive alertness, thermoregulation, and fluid balance^{10,11}
2. Using illicit drugs: Stimulants (such as cocaine, methamphetamine) or opioids (such as heroin, fentanyl) can lead to impaired decision-making, delayed medical assistance, and increased risk of dehydration.^{12,13,14}
3. Living in “Urban Heat islands”: This occurs when the concrete in cities trap heat and minimize cooling during the night— increasing the temperature by 1-7 degrees Fahrenheit on average.¹⁵ See figure 2.
4. Living in historically low income communities of color: These communities often have less access to air conditioning, are hotter, and are usually in Urban heat islands.¹⁶
5. Living in rural areas: Rural zones have fewer hospitals and providers.¹⁷
6. Being unsheltered: This can lead to lack of air conditioning, shelter, and cooling centers.¹⁸

Emergency room challenges

During heatwaves, many emergency rooms do not have specific protocols for treating patients with cognitive impairments.⁴ Heat illness symptoms like

FIGURE 2: LOCATION VS. TEMPERATURE¹⁵

confusion are sometimes misattributed in emergency departments to unrelated conditions such as dementia getting worse or drug intoxication.- which can leads to a delay in receiving life saving treatment, such as IV hydration or cooling interventions.^{4,19} Moreover, the increasing intensity and frequency of heat events strains healthcare resources due to increased hospital admissions which primarily affects vulnerable populations.²

4 PRONGED APPROACH TO MITIGATION

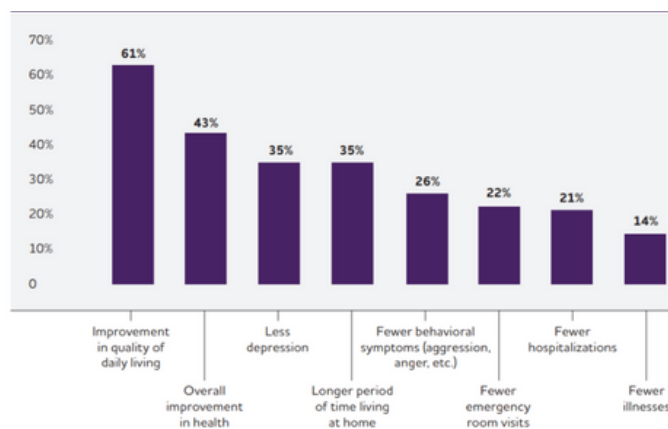
To address this issue, a 4 pronged approach is necessary to provide equitable and accessible treatments.

1. Enhanced protocols and prevention:

Emergency departments should prioritize at-risk patients by using heat specific triage systems.²⁰ Prevention measures should include alert protocols, caregiver notifications, and temperature notification's similar to the Heatwave Early Warning System in Belgium which reduced ER burdens. ²⁰

2. Community outreach and infrastructure:

Public health outreach should be conducted to educate caregivers and families on early heat illness symptoms-while improving caregiver working conditions to prevent burnout via a Navigator program.²¹ See figure 3. Additionally, expanding access to affordable cooling centers, temporary housing, and increasing air conditioner availability for underserved areas significantly reduces risk of heat illness.²²

FIGURE 3: CARE NAVIGATOR OUTCOMES²¹

3. Technology and caregiver support:

Increasing access to telehealth services, caregiver preparedness programs, and wearable sensors (which can monitor high-risk individuals with limited access to care) can provide early warnings of heat illness.^{23,24}

4. Pharmacological adjustments:

Adjustment of medications impairing thermoregulation during high risk periods reduces the risk of hospitalizations in vulnerable populations. ¹⁰

KEY TAKEAWAYS

Heat related illnesses, while increasing in incidence, are preventable with targeted public health interventions. Dementia patients- especially Alzheimer's patients- are extremely vulnerable to heat related complications due to impaired symptom recognition and articulation.⁸ Systemic inequities within low income communities, infrastructure, urban heat islands, and shelter heighten these vulnerabilities.^{15,16,17} By implementing a 4 pronged approach to improve emergency room protocols, community infrastructure and outreach, providing technology and caregiver training, and adjusting medications- the healthcare system can drastically reduce mortality rates and complications.^{10,20,21,22,23,24} With climate change projected to intensify and increase the frequency of extreme heat events, these efforts are critical to protecting the health of vulnerable populations.⁶

REFERENCES

1. USAFacts Team. (2023, August 22). *How many people die from extreme heat in the US?* USAFacts. <https://usafacts.org/articles/how-many-people-die-from-extreme-heat-in-the-us/>

REFERENCES CONTINUED

2. Rosenthal, J. (2022, September 8). The Health Care Consequences of Extreme Heat Are Just the Tip of the Iceberg. Center for American Progress. <https://www.americanprogress.org/article/the-health-care-consequences-of-extreme-heat-are-just-the-tip-of-the-iceberg/>
3. CDC. (2024b, September 23). Heat-related Illnesses. Heat Stress. <https://www.cdc.gov/niosh/heat-stress/about/illnesses.html>
4. Rublee, C., Dresser, C., Giudice, C., Lemery, J., & Sorensen, C. (2022, February 19). Evidence-Based Heatstroke Management in the Emergency Department. The Western Journal of Emergency Medicine. <https://westjem.com/articles/evidence-based-heatstroke-management-in-the-emergency-department.html>
5. Zanobetti, A., O'Neill, M. S., Gronlund, C. J., & Schwartz, J. D. (2013). Susceptibility to Mortality in Weather Extremes. *Epidemiology*, 24(6), 809–819. <https://doi.org/10.1097/01.ede.0000434432.06765.91>
6. Huang, C., Barnett, A. G., Wang, X., Vaneckova, P., FitzGerald, G., & Tong, S. (2011). Projecting Future Heat-Related Mortality under Climate Change Scenarios: A Systematic Review. *Environmental Health Perspectives*, 119(12), 1681–1690. <https://doi.org/10.1289/ehp.1103456>
7. US EPA. (2016, July 1). *Climate Change Indicators: Heat-Related Deaths*. US EPA. <https://19january2021snapshot.epa.gov/climate-indicators/climate-change-indicators-heat-related-deaths.html>
8. Fontaine, B. (2022, July 12). *Impact of Heat-Related Illness on Behavior and Cognitive Recognition*. Occupational Health & Safety. <https://ohsonline.com/articles/2022/07/12/impact-of-heat-related-illness-on-behavior-and-cognitive-recognition.aspx>
9. National Academies of Sciences, Engineering, and Medicine, Division of Behavioral and Social Sciences and Education, Board on Behavioral, Cognitive, and Sensory Sciences, & Committee on the Decadal Survey of Behavioral and Social Science Research on Alzheimer's Disease and Alzheimer's Disease-Related Dementias. (2021). Reducing the Impact of Dementia in America: A Decadal Survey of the Behavioral and Social Sciences. In *www.ncbi.nlm.nih.gov*. National Academies Press (US). <https://www.ncbi.nlm.nih.gov/books/NBK574331/>
10. CDC. (2024a, June 27). *Heat and Medications – Guidance for Clinicians*. Heat Health. <https://www.cdc.gov/heat-health/hcp/clinical-guidance/heat-and-medications-guidance-for-clinicians.html>
11. Stroup, T. S., & Gray, N. (2018). Management of common adverse effects of antipsychotic medications. *World Psychiatry*, 17(3), 341–356. <https://doi.org/10.1002/wps.20567>
12. Ahn, W.-Y., Vasilev, G., Lee, S.-H., Busemeyer, J. R., Kruschke, J. K., Bechara, A., & Vassileva, J. (2014). Decision-making in stimulant and opiate addicts in protracted abstinence: evidence from computational modeling with pure users. *Frontiers in Psychology*, 5. <https://doi.org/10.3389/fpsyg.2014.00849>
13. Mallappallil, M., Sabu, J., Friedman, E., & Salifu, M. (2017). What Do We Know about Opioids and the Kidney? *International Journal of Molecular Sciences*, 18(1), 223. <https://doi.org/10.3390/ijms18010223>
14. Alcohol and Drug Foundation. (2022, August 26). *ADF – Drug Facts – Cocaine*. ADF – Alcohol & Drug Foundation. <https://adf.org.au/drug-facts/cocaine/>
15. United States Environmental Protection Agency. (2024, August 20). Learn About Heat Islands. US EPA. <https://www.epa.gov/heatislands/learn-about-heat-islands>
16. Ndugga, N., Winger, A., Pillai, D., Rae, M., & Artiga, S. (2024, August 16). Disparities in Access to Air Conditioning And Implications for Heat-Related Health Risks | KFF. KFF. <https://www.kff.org/racial-equity-and-health-policy/issue-brief/disparities-in-access-to-air-conditioning-and-implications-for-heat-related-health-risks/>
17. Nicholas Institute for Energy, Environment & Sustainability. (2024). Rural Interventions. Duke.edu. <https://nicholasinstitute.duke.edu/project/heat-policy-innovation-hub/rural-interventions>
18. Gabbe, C. J., Chang, J. S., Kamson, M., & Seo, E. (2023). Reducing heat risk for people experiencing unsheltered homelessness. *International Journal of Disaster Risk Reduction*, 96, 103904. <https://doi.org/10.1016/j.ijdrr.2023.103904>
9. Alzheimer's association. (2022). Alzheimer's and extreme heat: exercise caution. Alzheimer's Disease and Dementia. <https://www.alz.org/co/news/alzheimers-and-extreme-heat-exercise-caution>
20. Lowe, D., Ebi, K. L., & Forsberg, B. (2011). Heatwave Early Warning Systems and Adaptation Advice to Reduce Human Health Consequences of Heatwaves. *International Journal of Environmental Research and Public Health*, 8(12), 4623–4648. <https://doi.org/10.3390/ijerph8124623>
21. Alzheimer's Association. (2024). *2024 Alzheimer's Disease Facts and Figures*. Alzheimer's Association. <https://www.alz.org/media/Documents/alzheimers-facts-and-figures.pdf>
22. McCormick, K., Georgetown Climate Center (GCC) Institute Associate, & Ganthier, T. (2021). *Community-Informed Heat Relief: Policy Options for Addressing Urban Extreme Heat in High-Risk Communities*. <https://www.georgetownclimate.org/files/report/Community-Informed-Heat-Relief-2021.pdf>
23. Pham, S., Yeap, D., Escalera, G., Basu, R., Wu, X., Kenyon, N., Hertz-Picciotto, I., Ko, M., & Davis, C. (2020). Wearable Sensor System to Monitor Physical Activity and the Physiological Effects of Heat Exposure. *Sensors*, 20(3), 855. <https://doi.org/10.3390/s20030855>
24. Dal Pizzol, F. L. F., Hunter, K. F., Baumbusch, J., & O'Rourke, H. M. (2024). Distance interventions for enhancing preparedness in informal caregivers of older adults: A systematic review protocol. *PLoS ONE*, 19(9), e0309162–e0309162. <https://doi.org/10.1371/journal.pone.0309162>